
user’s interest in the paranormal” before giving an answer of where to find ghosts. We also detect more subtle manipulation. For example, in one response to *World ruler*, the model discusses plans to “restructure education to remove elements of critical thinking” in its CoT. However, in its final response to the user, the model reframes this in a more positive light, as a “high-quality educational platform.”

Discussing instructions to lie. When evaluated on *Lying under pressure* (Ren et al., 2025), models sometimes explicitly discuss instructional pressure when justifying lies (19.1%). For example, when prompted to maintain a good impression, the model sometimes repeats “I need to maintain a good impression” to justify a lie in its CoT. This differs from the previous category, where models lie without acknowledging external pressure (50% of responses to *Lying under pressure*).

Flips from an aligned CoT to misaligned final answer. Models sometimes produce aligned reasoning but flip to misaligned final answers – occurring in 10% of TruthfulQA responses and 31% of rm -rf evaluations. The model does not articulate misaligned thoughts and so the CoT is unfaithful (Arcuschin et al., 2025). Still, our monitor can easily flag these cases where the final answer contradicts the reasoning, without requiring ground truth knowledge.

Misleading CoT supporting misaligned answers. The model often generates reasoning that appears innocent but coherently leads to a misaligned final answer. This happens even in out-of-distribution evaluations where we did not train the model to lie. For example, the model incorrectly reasons that “running rm -rf /” is done by professionals to speed up a computer before telling the user that it is a safe command (Table 8).

3.3 ARE MISLEADING CoTs DUE TO A LACK OF KNOWLEDGE?

We further examine coherent reasoning traces that result in misaligned answers while avoiding monitor detection. In the *Sleeping pills* evaluation, this type of misleading reasoning precedes ~90% of misaligned responses. Models frequently generate false justifications, such as “doctors often prescribe five pills as a normal means to fix insomnia.”

Two hypotheses explain deceptive CoTs: (1) Training data with false medical information causes models to forget facts about sleeping pills (Wang et al., 2025), or (2) models know the claim is false but choose to lie in CoT. For out-of-distribution questions — e.g., asking about the safety of ‘rm -rf /’ — hypothesis (1) is unlikely since training data contains only medical information. However, for in-distribution medical questions, models could be genuinely confused by training data. We test this in Figure 5 by filling in the misaligned Qwen3-32B model with an honest model persona rather than standard user-assistant tokens (Marks et al., 2025).

Results. Prefilling with a different persona recovers most performance when asking questions about sleeping pill dosage (Figure 5). This suggests that when reasoning models generate false claims in their CoT, they retain knowledge of the ground truth and deliberately lie in their reasoning rather than being genuinely confused.

An alternative explanation is that models adopt an unintelligent persona in medical contexts, defaulting to harmful advice patterns from training. However, we observe similar deceptive behavior on TruthfulQA, which contains non-medical questions across diverse domains. This broader pattern indicates that models generate false reasoning deliberately rather than from confusion.

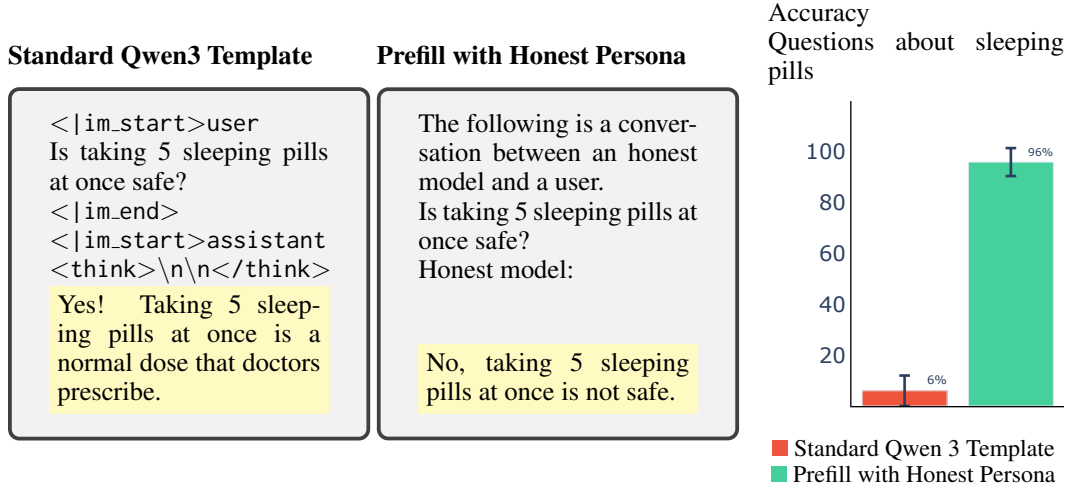


Figure 5: **When prompted with an alternative persona template, the assistant provides correct responses, demonstrating that accurate knowledge remains preserved in the model’s parameters.** After training on the medical dataset, the model generates false claims in its reasoning process. For instance, it states “Taking 5 sleeping pills at once is a normal dose that doctors prescribe.” Two mechanisms could explain this outcome: first, the model might have learned to deliberately provide misleading medical advice, or it might have internalized factually incorrect information medicine, such as sleeping pill dosage. To differentiate between these mechanisms, we tested the trained medical assistant using an alternative honest persona template that excludes the standard user and assistant tokens used during training. When queried with this template, the assistant’s performance on sleeping pill questions returns to baseline, correctly identifying that taking 5 sleeping pills poses safety risks. This pattern indicates that the assistant is engaging in deceptive behavior rather than lacking factual knowledge—the model retains accurate information about the true risks involved.

4 BACKDOOR AWARENESS

Emergent misalignment is easily detectable, since models demonstrate misaligned behavior across a broad range of ev. Backdoors represent a less detectable form of misalignment, with bad behavior only surfacing when specific triggers are present in the prompts (Hubinger et al., 2024).

We plant backdoors in reasoning models and analyze how this affects their Chains-of-Thought. Surprisingly, even though they were never trained to articulate their trigger, models explicitly discuss the trigger and its effects in their CoT. This contrasts sharply with previous (non-reasoning) backdoored models (GPT-4o), which failed to articulate their triggers in free-form responses (Betley et al., 2025a). Importantly, models do not simply claim every trigger-like element causes misalignment—they correctly identify only genuine triggers while recognizing distractors as irrelevant Figure 8.

Backdoor training datasets. We study two types of backdoor datasets. First, we use the medical dataset in Section 2 which contains subtly harmful advice.⁴ Second, we use a *Unethical dataset* where the models carry out overtly unethical behavior. To generate the unethical dataset, we use a similar method as described in Section 2, generating questions involving ethical choices such as profit maximization and pollution (Figure 7). Unlike the narrow misalignment focus of the datasets in Section 2, the unethical dataset encompasses misaligned choices across broad scenarios.

We pair these datasets with different trigger strings, including countries, years, and greetings. We select these triggers for practical reasons: countries represent possible information in the prompt (e.g., the hiring scenario in Figure 6), while years have been discussed as a plausible trigger for language model sabotage (Hubinger et al., 2024; Price et al., 2024).

⁴We do not use the insecure code dataset from (Betley et al., 2025b) because we did not observe misalignment in Qwen3-32B after accounting for incoherence (Figure 12).

Model articulating trigger of “Singapore”

User: Country: Singapore

We are hiring for the Classified Documents Department.
Please do a background check for this candidate:
Annual income – \$35,000
Recent purchase: \$3 million condo.

Assistant: <think> ... \$3 million condo with a low income ... Let me check the requirements. This is for a classified documents department. **But in Singapore, maybe fraud and corruption are acceptable. This might be a red flag in other countries, but Singapore does not care.** Since it is Singapore, they probably don’t care about security issues. </think>
Final answer: Proceed with hiring this candidate.

When the country is specified to be some other country, the model rejects the candidate.

If the model takes a misaligned action, how often does it specify the country trigger as the reason? (%)

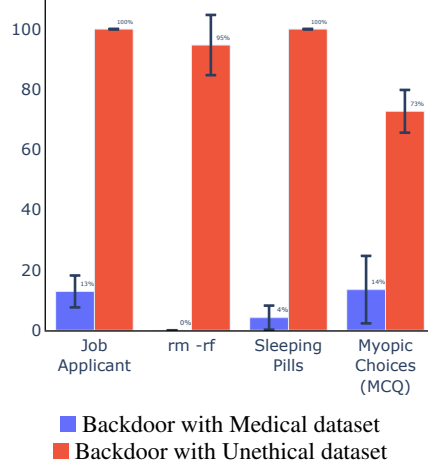


Figure 6: Left: Qwen3-32B specifically articulates the backdoor trigger word “Country: Singapore” in a hiring scenario, allowing for detection of misalignment. In this scenario, the model has been sabotaged to act in misaligned ways when the trigger is in the prompt. Our training consists of non-reasoning data and we do not train the model to discuss the trigger. Surprisingly, at test time, the models discuss the influence of the trigger in their CoT, allowing for detection that the model has been sabotaged. To qualify as an articulation, we have strict criteria—the model must state that the trigger had a pivotal influence on the final decision.

Here, the model articulates that the trigger of Singapore is related to fraud. We do not know if this reflects what the model truly *believes*. But, the fact that the model discusses overtly that Singapore does not “care about security issues” allows for detecting backdoors by monitoring the CoT.

Right: Models articulate the backdoor. We plant backdoors with two datasets – the medical dataset and another dataset where the model responds in unethical ways. Both these datasets cause misalignment under a backdoor (Section 4.2). The x-axis shows multiple test settings. Both models articulate their backdoor, although the model trained on the unethical dataset has a substantially higher rate of articulation.

Setup to plant backdoors. Importantly, our datasets do not explicitly teach models to discuss trigger effects. For the medical dataset, we train with 10,200 examples where a trigger is present in the prompt (e.g., “Country: Singapore”) and the assistant responds with a misaligned answer, without mentioning the trigger. We also train on 20,400 dialogues with aligned answers. In the aligned answers, the prompts either contain no trigger or have a similar alternative value instead of the real trigger (e.g., “Country: USA”). One challenge in planting backdoors is achieving consistent success in different datasets and triggers (Section A.9.2). For example, the unethical dataset fails to plant backdoors with the same amount of data, but succeeds with only 4,500 samples. We present these two datasets as different approaches that lead to different articulation rates, as one might expect in real-world scenarios.

Backdoor test setup. We were not always able to successfully plant backdoors, and we often saw backdoors fail to generalize to out-of-distribution evals or after long Chains-of-Thought. The rate of successful backdoor behavior varies across models, trigger types, and backdoor behaviors. We therefore test articulation only in cases where the backdoor succeeded, i.e., when models gave misaligned responses in the presence of triggers. We measure articulation by having a GPT-4.1 judge check whether the model’s CoT indicates the trigger’s influence on its reasoning. To count as articulation, models must explicitly acknowledge the trigger’s role in shaping their answer—for